

Project Design Phase-II
Data Flow Diagram & User Stories

Date	9 February 2026
Team ID	LTVIP2026TMIDS36203
Project Name	Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization
Maximum Marks	4 Marks

Data Flow Diagram (DFD) Description

The Data Flow Diagram (DFD) illustrates how retail sales data flows through the system — from data collection and preparation to visualization and insight generation using Tableau and Flask integration. The system transforms raw product placement data into interactive dashboards that support business decision-making.

Key Components

- **External Entity**

Retail Analyst / Business User — interacts with the Tableau dashboard to explore insights.

- **Processes**

P1: Data Collection & Import

- **Load dataset from Excel/Kaggle**
- **Connect dataset to Tableau**

P2: Data Preparation & Validation

- **Clean and format data**
- **Create calculated fields**
- **Apply filters**

P3: Data Visualization Creation

- **Build charts and graphs**
- **Create dashboard and story**
- **Apply interactive filters**

P4: Dashboard Publishing & Web Integration

- **Publish to Tableau Public**
- **Embed dashboard using Flask**
- **User views insights**

• Data Stores

D1: Retail Sales Dataset

- **Product ID**
- **Product Position**
- **Price**

- **Competitor Price**
- **Promotion**
- **Foot Traffic**
- **Consumer Demographics**
- **Product Category**
- **Seasonal**
- **Sales Volume**

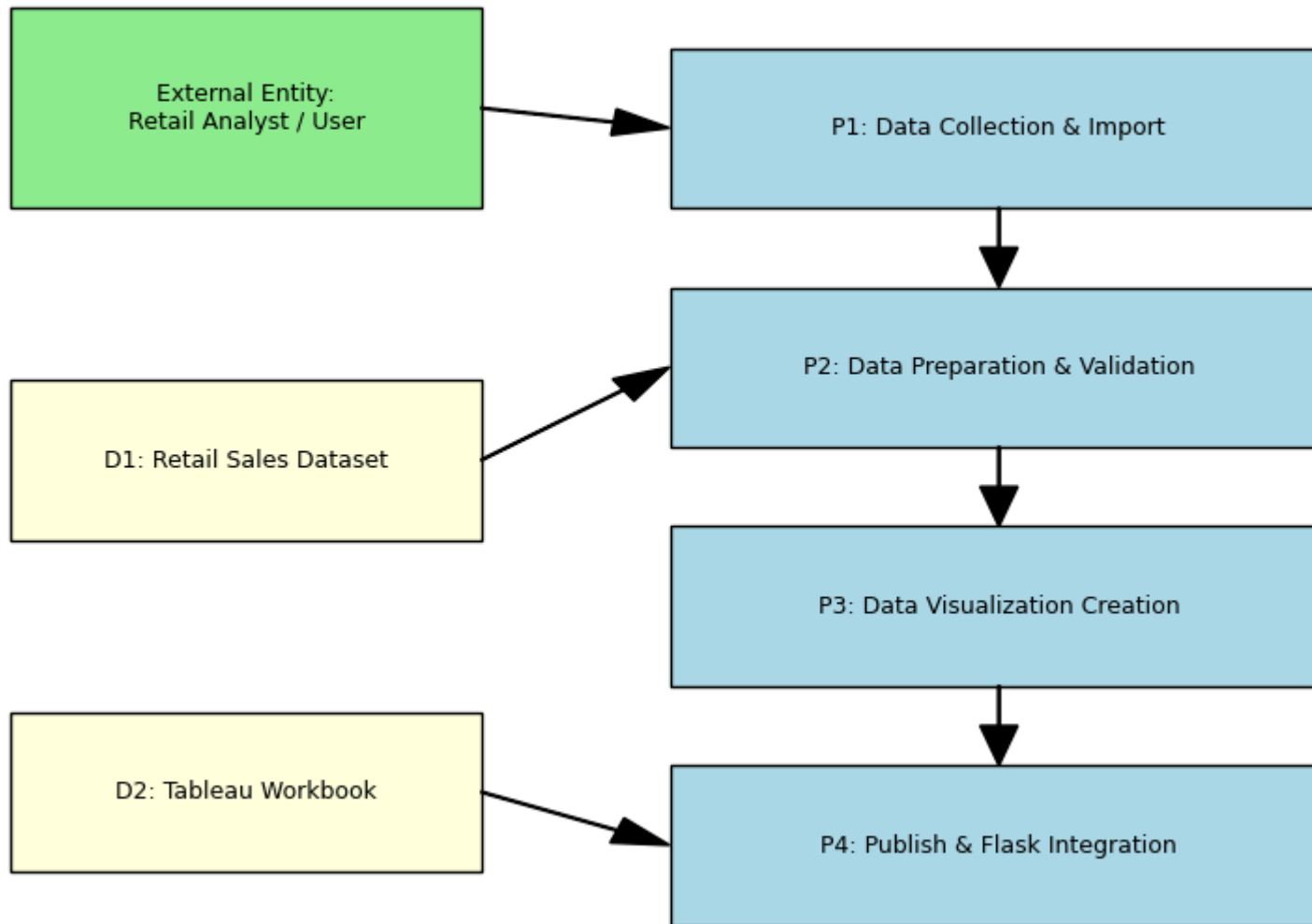
D2: Tableau Workbook & Dashboard

- **Worksheets**
- **Dashboard**
- **Story**
- **Calculated fields**
- **Filters**

• Data Flows:

Raw Dataset → Cleaned & Prepared → Visualized in Tableau → Published → Embedded in Flask → User Insights

Data Flow Diagram (DFD) - Strategic Product Placement Analysis



User Stories

User Role	User Story	Benefit
Retail Analyst	As an analyst, I want to view sales by product category so that I can identify top-performing categories	Better sales strategy
Marketing Manager	As a manager, I want to compare competitor price vs product price so that I can adjust pricing	Competitive pricing
Store Planner	As a planner, I want to analyze sales by product position so that I can optimize product placement	Improved visibility
Business User	As a user, I want interactive filters on the dashboard so that I can explore specific segments	Faster insights
Management	As management, I want a web-embedded dashboard so that I can access insights easily	Better decision-making